ResearchPaL

Multi-Agent Academic RAG Assistant with Layout-Aware Parsing

Executive Summary: ResearchPaL is a production-grade multi-agent research assistant designed to solve the critical drawbacks of conventional PDF Question-Answering systems. Standard RAG pipelines treat scientific PDFs as flat, unformatted plain text, discarding vital information stored in **mathematical formulas, structured tables, visual figures, and page hierarchies**. ResearchPaL preserves complete document topology via layout-aware multi-modal parsing, hybrid dense-sparse retrieval (RRF), cross-encoder reranking, and a stateful LangGraph agent graph with deterministic page-level citation verification.

1. Technology Stack Architecture

The system employs a decoupled, modular architecture emphasizing speed, client-side privacy (BYOK), and resilient multi-model failovers:

Layer	Technologies & Frameworks	Key Role & Implementation
Frontend UI	React 18, Vite, Lucide Icons, KaTeX, Tailwind / Custom Dark Glassmorphism	Responsive workspace with discovery feed, real-time response streaming, clickable citation badge scrolling, and full-screen figure lightbox zoom.
Backend API	Python 3.10+, FastAPI, Uvicorn, Pydantic v2, HTTPX	Async REST endpoints, Server-Sent streaming (StreamingResponse), background worker thread pool, ArXiv URL rewriting, and magic-byte validation.
Multi-Agent Graph	LangGraph, LangChain Core	Stateful DAG workflow executing: Intent classification, dynamic context retrieval, cross-encoder reranking, and citation-guarded generation.
Vector Store & DB	Supabase (PostgreSQL 15+, pgvector, Storage Buckets)	Unified relational storage + vector indexing; 768-dimensional cosine index, PostgreSQL array filtering for multi-doc queries, public figure storage.
Document Ingestion	PyMuPDF (fitz), Adobe PDF Services API, Gemini Vision / Groq	Dual-mode parser: Basic Mode (fast text extraction) and Advanced Mode (layout-aware JSON structure, tables, and visual diagram bounding boxes).
Embeddings & Reranker	Jina Embeddings v3 (768d), Gemini Embedding, Jina Reranker v2	Dense semantic vectorization and cross-encoder reranking (refining top-15 retrieved candidates down to top-5 high-signal chunks).
LLM Providers & Failover	Google Gemini 1.5/2.5/Flash-Lite, OpenAI GPT-4o-mini, Anthropic Claude 3.5, Groq Llama 3.2 Vision	Multi-provider flexibility; automated zero-downtime failover to Groq vision/chat upon encountering Gemini 429 quota exhaustion.
Security & Privacy	Bring-Your-Own-Key (BYOK), Browser localStorage	Zero server-side persistence of user AI credentials; keys passed ephemerally via request headers and never written to database tables.

2. Core Functionalities & System Capabilities

ResearchPaL delivers advanced research synthesis through eight specialized engineering capabilities:

1. Multi-Modal Layout-Aware Parsing: Unlike naive PDF splitters that discard non-text elements, the ingestion pipeline preserves tables, formulas, and visual figures. In Advanced Mode, Adobe PDF Services parses the document layout tree into structured content blocks, while Gemini/Groq Vision generates natural language semantic summaries of every diagram.

2. Semantic Text Chunking with Cosine Windowing: Rather than rigid fixed-character boundaries that sever sentences, ResearchPaL groups text into sentences, computes batch embeddings, and evaluates cosine similarity between adjacent sentences. Chunks only break at natural semantic boundaries (similarity < 0.75), maximizing contextual cohesion.

3. Hybrid Search with Reciprocal Rank Fusion (RRF): Combines dense semantic vector retrieval (Jina/Gemini embeddings) with keyword term matching. Merged using Reciprocal Rank Fusion: Score(d) = 1/(60 + Rank_vector) + 1/(60 + Rank_keyword). This completely eliminates terminology mismatch on exact acronyms and equation names.

4. Cross-Encoder Reranking (Jina Reranker v2): High-recall initial retrieval (k=15) is funneled through a cross-encoder model to compute deep query-passage attention. This delivers a +4.2% Context Precision boost, scoring figure and table metadata directly against user intent before generation.

5. Multi-Paper Literature Review: Users can select multiple indexed papers from the library sidebar. The backend queries across composite document IDs in PostgreSQL, synthesizing comparative answers with multi-document attribution.

6. Grounded Responses & Hallucination Guard: Every assertion generated by the LLM is anchored by explicit page-level citations (e.g., [Source 1]). An automated post-processing validator scrubs phantom citation indexes that exceed retrieved evidence limits.

7. Paper Discovery & Direct ArXiv Ingestion: Integrated with the Papers With Code and ArXiv APIs. Users can search trending research or paste abstract URLs (e.g. arxiv.org/abs/...), which are automatically translated to direct PDF streams and indexed with magic-byte (%PDF) validation.

8. Interactive Evidence Inspection & Lightbox Zoom: Clicking a citation badge in the chat dynamically scrolls to and highlights the exact evidence card. Extracted figures and tables can be clicked to open in an ultra-crisp full-resolution modal lightbox.

3. LangGraph Multi-Agent Architecture

Retrieval and generation are governed by a stateful Directed Acyclic Graph (DAG) built with LangGraph:

Node / Agent	Function & Trigger Condition	Output Artifact
Intent Router	Classifies user query into 4 distinct intents: greetings (fast chat), visual_tabular (prioritize figures/tables), global_synth (broad overview), or doc_search.	State variable: route_decision directing graph execution path.
Hybrid Retriever	Executes parallel dense pgvector similarity + keyword search across selected document IDs; merges candidate lists via Reciprocal Rank Fusion.	Top-15 candidate document chunks (text, tables, image summaries).
Jina Reranker	Applies cross-encoder scoring between user prompt and candidate chunks; filters low-relevance noise down to the top 5 most salient contexts.	Reranked top-5 context chunks with relevance score metrics.
Response Generator	Streams synthesized answer using conversational history, inserting grounded [Source N] markers and appending structured source JSON metadata.	Real-time token stream + verified sources metadata payload.

4. End-to-End User Flow & Operational Journey

The interaction workflow is designed for frictionless research exploration across four phases:

Phase	User Action	System Execution & Background Operations
Phase 1: Configuration	1. Open Web Application. 2. Click Settings & enter API keys (Gemini, OpenAI, Anthropic, or Jina).	Keys are validated and saved exclusively to browser localStorage (zero database logging).
Phase 2: Paper Ingestion	**Option A:** Upload local PDF file. **Option B:** Paste ArXiv URL or click Trending Papers.	1. Backend validates magic bytes (%PDF). 2. Dispatches async ingestion task in FastAPI background thread. 3. PyMuPDF or Adobe Extract parses text, tables & figure bounding boxes. 4. Semantic Chunking breaks text into cohesive paragraphs. 5. Gemini/Groq Vision summarizes visual figures. 6. Jina/Gemini generates 768d embeddings stored in Supabase pgvector. 7. Real-time notification updates library list upon indexing completion.
Phase 3: Workspace Setup	Select single paper or multi-select multiple papers from Library Sidebar.	Frontend configures active context scope (document_ids) for targeted queries or comparative literature cross-analysis.
Phase 4: Interactive Synthesis	Type research questions (e.g. *'Compare ablation results in Table 2 with Figure 4'*).	1. Intent Router identifies visual/tabular and comparative requirements. 2. Hybrid search fetches top candidates across all selected papers. 3. Jina Reranker scores candidates and isolates top 5 context passages. 4. Streaming response renders markdown, KaTeX math equations, and citations. 5. Clicking [Source 1] scrolls to and expands the page excerpt. 6. Clicking any figure thumbnail launches high-res Lightbox zoom modal.

5. Empirical Performance & Benchmark Results

Evaluated on a curated academic benchmark across 4 research papers with 10 multi-hop questions each:

Benchmark Metric	Score	Benchmark Configuration	Engineering Significance
Context Recall	**0.927**	Chunk=2500, k=13	Retrieves 92.7% of all ground-truth evidence required for answers.
Context Precision	**0.390**	Hybrid + Jina Reranker	+4.2% precision boost over standard hybrid search without reranking.
MRR (Mean Reciprocal Rank)	**0.757**	Chunk=2500	First relevant chunk is positioned in top 1-2 search slots on average.
Hit@5	**1.000**	Chunk=2500	100% of all queries locate relevant evidence within the top 5 results.
Figure / Table Hit@1	**1.000**	k ≥ 13 (Advanced Mode)	Perfect retrieval accuracy for visual diagrams and complex table queries.
P50 Query Latency	**646 ms**	Default text mode	Sub-second vector retrieval and rank fusion processing.
P50 Response Latency	**2,620 ms**	Full pipeline end-to-end	Rapid first-token generation and streaming response delivery.